

# It from Fix: The Kernel Principle

Stationarity in time, scale, and structure: a working note with toy models, controls, and negative results

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## Abstract

Three familiar statements share one algebraic form. The Wheeler-DeWitt constraint  $\hat{H}|\Psi\rangle = 0$  says the quantum state of the universe is stationary in time. The fixed-point condition  $\beta(g_*) = 0$  of asymptotic safety says fundamental couplings are stationary in scale. Conservation of a structural kernel under an update map,  $\text{Ad}_U(K) = K$ , says structure is stationary under discrete evolution. I propose to read these as instances of a single selection principle: the physically realized objects are elements of  $\ker(\mathcal{G})$  for the relevant flow generator  $\mathcal{G}$ , and observed dynamics is relational flow around them. This note states the principle, is explicit about what it does and does not predict, and reports the computational program built around it. That program includes reproductions of published asymptotic-safety results, a family of stationary spectral toy models in which chirality produces complex projected scaling exponents, and a series of controlled attempts to make gravitational attraction emerge from persistence and information exchange in relational graphs. The attempts failed under fair controls, and I report the failures with the same care as the constructions that worked. The note ends with the criteria a future positive result would have to meet.

## 1 Introduction

Physics textbooks present time evolution, scale dependence, and conservation laws as separate chapters. This note explores the possibility that, at the fundamental level, the three chapters are one statement, and that the statement is not an evolution equation but a constraint: stationarity.

The parts of the argument are standard. Canonical quantum gravity has long known that the Hamiltonian of a closed universe annihilates the physical state [1], with experienced time emerging relationally [2]. The asymptotic-safety program proposes that couplings are coordinates of a renormalization-group fixed point rather than free parameters [3, 4]. What this note adds is modest and I want to be upfront about it: a reading that treats these, together with structural conservation laws of the kind appearing in pregeometric approaches, as one selection rule; a set of toy models probing whether the reading can be implemented; and an experimental record, including negative results, of an attempt to push the reading toward emergent gravity.

A word on method. Much of this note is a lab notebook as much as a theory paper. Where a result reproduces published work, I say so. Where a construction is elementary, I say so. Where an experiment failed, the failure is reported with its controls, because in this corner of theory space the failures carry most of the information.

## 2 Three stationarities

**Time.** In canonical general relativity the Hamiltonian is a constraint. Upon quantization, the physical state satisfies the Wheeler-DeWitt equation

$$\hat{H} |\Psi\rangle = 0. \quad (1)$$

So  $|\Psi\rangle$  is an exact fixed point of the semigroup generated by  $\hat{H}$ : for all  $\tau$ ,  $e^{-\tau\hat{H}}|\Psi\rangle = |\Psi\rangle$ . The universe, described globally, does not evolve. Experienced time arises from correlations between subsystems, one of which serves as a clock [2]. Only relational, non-uniform drifts are empirical.

**Scale.** In the functional renormalization group, dimensionless couplings  $g_i(k)$  run with the coarse-graining scale  $k$  according to  $\partial_t g_i = \beta_i(g)$ ,  $t = \ln k$ . The asymptotic-safety hypothesis [3] is that the ultraviolet limit is controlled by a non-Gaussian fixed point

$$\beta_i(g_*) = 0, \quad (2)$$

so the observed constants are coordinates of a stationary point of the scale flow, approached along a finite-dimensional critical surface.

**Structure.** In pregeometric approaches, the primitive object is not a spacetime field but a kernel  $K$  preserved by an update map  $U$ :

$$\text{Ad}_U(K) \equiv U^\dagger K U = K. \quad (3)$$

Conservation of structure is stationarity under the discrete flow generated by  $U$ .

## 3 The kernel principle, and what it does not predict

**Principle 1** (Kernel Principle). *Let  $\mathcal{G}$  be the generator of a physical flow (time translation, scale translation, or structural update). The fundamental objects of the theory are elements of  $\ker(\mathcal{G})$ . Observed dynamics is the relational flow of subsystem correlations in a neighborhood of  $\ker(\mathcal{G})$ ; it is emergent, not fundamental.*

Equations (1), (2) and (3) are then one statement,  $\mathcal{G}(X) = 0$ , with

$$(\mathcal{G}, X) \in \left\{ (\hat{H}, |\Psi\rangle), (\beta \cdot \partial_g, g_*), (\text{Ad}_U - \text{id}, K) \right\}.$$

A note on terminology. We use *kernel* in the extended sense of the zero set of the generator. This coincides with the linear-algebra kernel when  $\mathcal{G}$  is linear, as for  $\hat{H}$  and  $\text{Ad}_U - \text{id}$ , and with the fixed points of the induced flow in general:  $X \in \ker(\mathcal{G})$  exactly when  $X$  is a fixed point of  $e^{-\tau\mathcal{G}}$  for all  $\tau$ . For the nonlinear beta function,  $\beta(g_*) = 0$  defines a zero of a vector field, the standard RG fixed point, and the kernel reading applies to the linearization around  $g_*$ . The principle is stated at the level of generators rather than flows because that is where the three equations share one form.

Two remarks sharpen the content, and a third limits it. First, the principle is not the trivial observation that stationary points exist in many theories; it is the claim that in each sector the *physically realized* object is the stationary one, seen through three different generators. Second, the principle inverts the usual explanatory order: stationarity is fundamental, and dynamics is what non-kernel degrees of freedom look like from inside.

The third remark is the honest one. In its present form the principle makes no prediction of its own. Every empirical statement in Section 8 (rigidity of constants, propagator softening, Lorentz recovery) is already a prediction of asymptotic safety or of any pregeometric emergence

program. Until one stationarity is derived from another, the kernel principle is a reading, not a theory. Section 10 states what a derivation would require. The rest of this note documents how far toy models currently get, which is: further than I expected on implementability, and nowhere yet on emergence.

### 3.1 Earlier quantitative tests within this program

The principle has already been tested quantitatively in one sector, by my earlier work on Standard Model couplings (the LVS program, 2024 to 2026, archived with this project), and the results constrain it in both directions. The strong variational form, in which the measured couplings would minimize a global stationarity action along the renormalization-group flow between the electroweak and Planck scales, is falsified: at one loop the optimum requires  $\alpha_s(M_Z) \approx 0.048$ , against the measured  $0.1179 \pm 0.0010$ , and the result is robust across three stationarity metrics. The weak form, stationarity imposed as a boundary condition at the Planck scale, requires gravitational anomalous dimensions  $f_g \approx 0.010$  and  $f_y \approx 0.013$ , while functional renormalization-group computations at the Reuter fixed point give  $f_g \approx 0.055$  and  $f_y \approx 0.004$  [18]. The discrepancy is a factor of three to five, within the roughly 60% truncation uncertainty of current computations, so the weak form is neither confirmed nor refuted, and its order of magnitude  $f \sim 10^{-2}$  follows from dimensional analysis alone, so it is not a distinctive signature. Any future version of the principle must be read against these two results: the selection rule does not operate as a global extremum over the Standard Model flow, and its boundary-condition form is not yet distinguishable from generic asymptotic safety.

## 4 Reproductions of known fixed-point results

This section is verification work. Everything in it reproduces published results; the value is independent confirmation and the fact that a single computer-algebra predicate, membership in  $\ker(\mathcal{G})$ , expresses all three stationarities. The computations start from the analytic beta functions of the Einstein-Hilbert truncation with optimized cutoff [5], transcribed from the published expressions, and from the published  $f(R)$  truncation data of [6].

### 4.1 Einstein-Hilbert truncation

In  $d = 4$ , with couplings rescaled as in [5], the simultaneous zeros of  $\beta_\lambda$  and  $\beta_g$  were located numerically (residual  $10^{-12}$ ) and then checked symbolically: the point

$$(\lambda_*, g_*) = \left(\frac{1}{4}, \frac{1}{64}\right) \quad (4)$$

annihilates both beta functions exactly as rational expressions, as found by Litim. The stability matrix yields the critical exponents

$$\theta_\pm = \frac{5 \pm i\sqrt{167}}{3} = 1.666667 \pm 4.307616 i, \quad (5)$$

matching the published analytic values to  $10^{-8}$ . The anomalous dimension at the fixed point is  $\eta_* = -2$  exactly. Both exponents have positive real part: the fixed point has two relevant directions forming a complex-conjugate pair, so renormalization-group trajectories spiral into it.

### 4.2 Continuous-dimension scan

Scanning  $d \in [2.6, 22]$  in steps of 0.2: the non-Gaussian fixed point exists on the whole range, the numerically located  $\lambda_*(d)$  agrees with the closed-form expression of [5] to  $10^{-10}$ , and the critical exponents are complex precisely for  $2.6 \leq d \leq 21.2$  at this resolution, closing near  $d \approx 21.4$  as predicted analytically. The spiral structure is not a four-dimensional accident.

### 4.3 $f(R)$ truncations

Using the published fixed-point data of [6] for polynomial truncations  $f(R) = \sum_{i=0}^n g_i R^i$ ,  $n = 1$  to 6: the fixed point survives every enlargement of the theory space, the complex pair survives in all truncations, and the dimension of the UV critical surface stabilizes at three from  $n \geq 2$ . The scheme-robust invariant  $\tilde{\Lambda}_* \tilde{G}_*$  lies between 0.11 and 0.13 for  $n \neq 2$ , consistent with exponential-cutoff computations [7].

## 5 Toy models: chirality and complex projected exponents

The question behind the toy models is narrow: can a single stationary kernel support both a relational clock sector and a scale sector whose projected flow spirals into a fixed point, the way the gravitational flow does? The answer is yes, and the mechanism is chirality. Before describing it I want to place it against prior work, because the mathematics involved is either elementary or already known, and the contribution is the dictionary, not the theorems.

### 5.1 The mechanism and its prior art

Decompose the effective generator obtained by projecting the kernel dynamics onto slow modes as  $L_K = S_K + \Omega_K$  with  $S_K$  symmetric and  $\Omega_K$  antisymmetric. For a projected  $2 \times 2$  operator

$$M = \begin{pmatrix} a & b + w \\ b - w & d \end{pmatrix},$$

the eigenvalues are complex exactly when

$$w^2 > \frac{(a - d)^2}{4} + b^2. \quad (6)$$

This is the discriminant of a  $2 \times 2$  matrix, elementary linear algebra. The statistical counterpart, that increasing the antisymmetric component of a random matrix raises the fraction of complex eigenvalues, is the subject of the real elliptic Ginibre ensemble in random matrix theory, where the interpolation between symmetric and asymmetric matrices and the statistics of real eigenvalues are studied in detail [8, 9]. And the physical interpretation of complex scaling exponents, log-periodic corrections and discrete scale invariance, has a substantial literature reviewed by Sornette [10]. None of (6), its ensemble version, or its interpretation is new.

What the toy models add is the assembly: a stationary kernel whose single spectrum provides, through two projections, a Page-Wootters clock sector and a spiraling scale sector, with the chirality of the kernel as the common origin of both the relational phases and the complex exponents.

### 5.2 Blind ensemble test

To check that the mechanism is not an artifact of construction, kernels were generated blind: random directed graphs with an orientation-strength parameter, no target exponents injected, and an automatic coarse-graining rule. The fraction of complex projected exponents rises from 0.000 at zero orientation to about 0.67 at orientation 1.0 and 0.80 at orientation 2.0 (Figure 1), and the association survives changes of graph size, density, and projection rule. Given the random-matrix background above, this behaves exactly as expected; the test is a sanity check that passed, not a discovery.

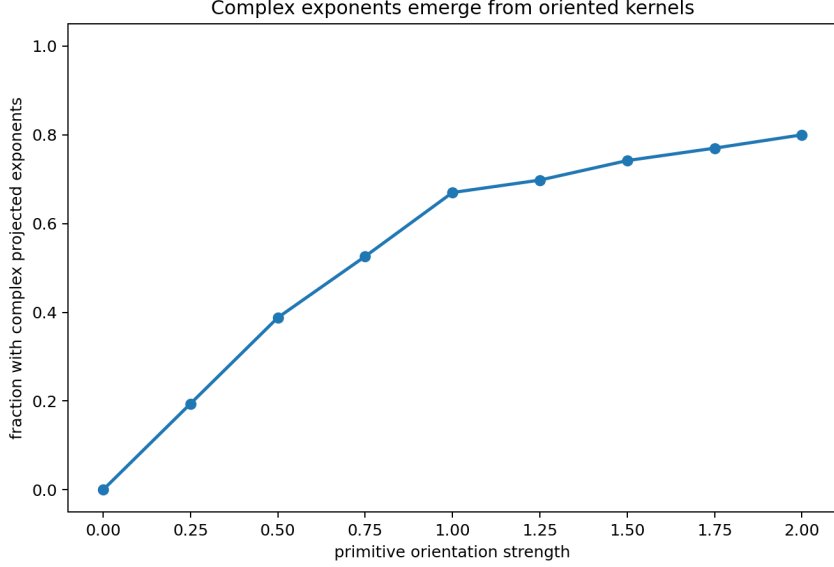


Figure 1: Blind ensemble test. Fraction of complex projected exponents as a function of the orientation strength of randomly generated kernels, with no target exponents injected. The rise is expected from random matrix theory; the test checks that the pipeline does not manufacture the signal.

### 5.3 A cautionary fit

A parameter search over chiral ring kernels (circulant structure, chiral and next-neighbor couplings) reaches a spiral ratio of 2.5845991 against the Einstein-Hilbert target  $\sqrt{167}/5 = 2.5845696$ , an absolute error of  $3 \times 10^{-5}$ . I report this as a warning rather than a result: a flexible enough ansatz family can match a single scalar to high precision. Without a held-out prediction, the match carries no evidential weight. A tetrahedral and an IVM-like polarized family were also searched; neither outperformed the ring, and perfectly symmetric tetrahedral constructions cancel the slow-mode chirality entirely.

## 6 Negative results: persistence does not make gravity

The second half of the program asked a more ambitious question. Suppose gravity is not fundamental but is the mechanism by which persistent fixed structures avoid dissolving back into the underlying relational space. Can attraction then emerge from local rules that only reward persistence or information exchange, in a graph with no coordinates, no metric, and no attraction term? This is close in spirit to quantum graphity [11] and to thermodynamic and entropic readings of gravity [12, 13], and the answer, in every unbiased version I tested, was no.

The experimental sequence, with emergent distances reconstructed from path costs only after the dynamics ran, was as follows.

| Rule                              | Mean distance response     | Verdict                    |
|-----------------------------------|----------------------------|----------------------------|
| Persistence-only reinforcement    | about $-3.3\%$ (repulsive) | negative                   |
| Random equal-budget halo          | $-2.6\%$                   | negative (control)         |
| Survival-optimized exchange       | about $-1\%$               | negative                   |
| Persistence + local observability | $-1.3\%$                   | negative                   |
| Hand-selected exchange halo       | $+10.3\%$                  | positive but hand-selected |
| Equal-budget targeted halo        | $+4.2\%$                   | encouraging, not emergent  |

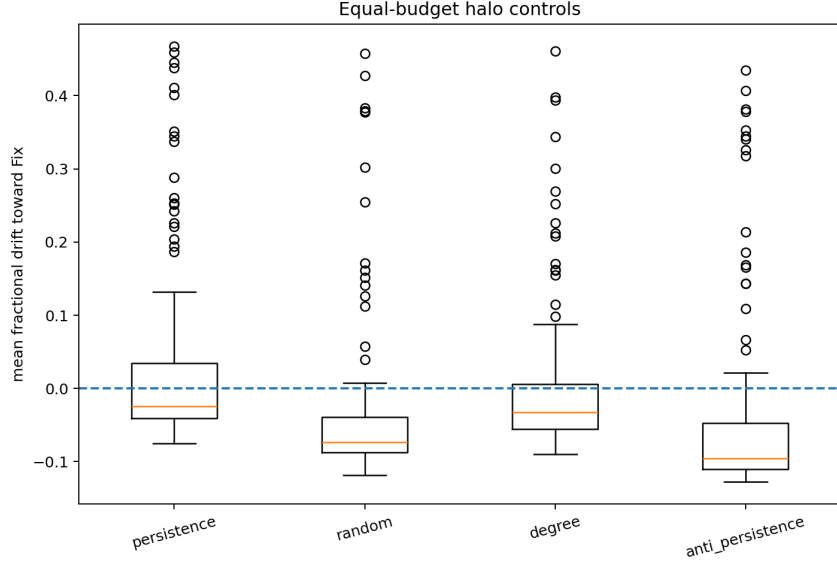


Figure 2: Equal-budget controls. All rules receive the same budget of new links; only the targeted exchange halo produces contraction, and it requires knowing which nodes to target. Rules that discover their exchanges by optimizing survival alone stay on the repulsive side.

(Sign convention: positive means contraction of the emergent distance between persistent structures, the attraction-like direction. Figure 2 shows the equal-budget comparison.)

Three lessons survived the controls. First, reinforcement that only consolidates a persistent core isolates it: the structure becomes more stable and more distant, the opposite of gravity. Second, an exchange halo around the core can reverse the sign, but in every positive run the halo was selected by hand or by a rule that already knew which nodes mattered; when the structure had to discover its exchanges by optimizing its own survival, the effect disappeared. Third, several early positive signals (joint persistence being cheaper, distances contracting) were reproduced by generic reinforcement in equal-budget controls and were therefore not specific to persistence at all. An earlier grid-based series found time-like signals (phase slowdown near high-traffic regions) but no deflection toward them, and one apparent drift was fully explained by hub accessibility in degree-matched null models.

I consider this table the most useful content of the note. It does not rule out emergence of gravity from persistence, but it rules out the easy versions, and it sharpens what a serious attempt must look like (Section 10).

These outcomes independently match my earlier modular-geometry program (RAPC, April 2026, archived with this project), in which iterated reinforcement flows on finite graphs produced either densification into a non-local mush or collapse to the empty graph unless sparsity was imposed through an information budget, and in which spontaneous bridging between fragmented geometric patches never occurred across any tested rule. Two toy programs built on different substrates, one modular and one relational, hitting the same walls suggests the obstacle is structural rather than an artifact of either construction.

## 7 Relation to the Weinberg-Witten theorem

If gravity is to emerge from a fixed-point structure rather than be fundamental, the Weinberg-Witten theorem [14] must be evaded. Its hypotheses are: a Lorentz-covariant conserved stress tensor, a graviton living in the same spacetime as the underlying theory, and exact Lorentz invariance. The kernel principle selects the pregeometric escape: if spacetime itself is emergent,

no fundamental local  $T_{\mu\nu}$  exists and the theorem does not apply. Holography exploits the same route. The corresponding obligation is severe: approximate locality, effective Lorentz covariance, and an emergent stress tensor must be recovered at low energy with observed precision. That obligation is the core open problem, shared with causal sets, group field theory, and tensor models.

## 8 Empirical status

The fixed-point picture makes contact with experiment in three ways. I stress again that these are commitments of asymptotic safety and of pregeometric emergence generally, not distinctive predictions of the kernel principle.

**Rigidity of constants.** If couplings sit on a fixed point, no secular drift of dimensionless constants is allowed beyond relational running. Optical clock comparisons with  $^{171}\text{Yb}^+$  constrain  $\dot{\alpha}/\alpha = 1.8(2.5) \times 10^{-19} \text{yr}^{-1}$ , consistent with zero [15]. The picture requires this null result and is falsified by any confirmed non-zero drift.

**Graviton propagator softening.** The fixed-point value  $\eta_* = -2$  modifies the graviton propagator at high momenta, with potential imprints on the primordial gravitational-wave spectrum, in the band of future space interferometry.

**Lorentz invariance.** Pregeometric emergence must reproduce Lorentz covariance at least to the level implied by the multimessenger constraint  $|v_{gw}/c - 1| \lesssim 10^{-15}$  from GW170817 [16]. Any framework in this family predicting residual violations above current bounds is excluded.

## 9 Limitations

The rational values  $(1/4, 1/64)$  are truncation and scheme artifacts; only the existence of the fixed point, the complex pair, the relevant-direction count, and approximately the invariant  $\tilde{\Lambda}_* \tilde{G}_*$  are expected to be universal. Robustness across  $f(R)$  truncations is robustness within one family of approximations. The toy-model mathematics of Section 5 is elementary or known; its role is architectural. The ring match is a fit. The projection rule that selects slow modes is chosen, not derived, and a physical theory must derive it. Emergent distances built from reinforced edges risk circularity when reinforcement also defines the dynamics; the controls of Section 6 were designed against exactly that, and they are the reason most of the positive signals died. Finally, no Lorentzian signature, light cone, or invariant causal speed has been derived anywhere in this program.

## 10 What a positive result would require

The kernel principle becomes a theory rather than a reading if the following can be shown, in order of increasing difficulty.

(i) *Derivation.* Exhibit a flow on kernel space whose stationarity condition reduces, in a coarse-grained limit, to a Wheeler-DeWitt-type constraint, and whose linearization around the kernel reproduces a renormalization-group flow with a non-Gaussian fixed point. This would derive one stationarity from another and give the principle content.

(ii) *Emergence.* A local update rule with no attraction term and no target gravitational number, preregistered before running, producing on held-out graph ensembles: universal attractive response across probe types, absence of the effect in degree- and accessibility-matched

null models, stable scaling with system size, and at least one quantitative relation (an exponent, a ratio) that was not fitted.

(iii) *Prediction*. At least one Standard Model dimensionless ratio from relevant-direction counting at a combined matter-gravity fixed point, in the spirit of [17].

Failure modes are explicit: a confirmed drift of  $\alpha$ , superluminal or dispersive gravitational-wave propagation, or the disappearance of the fixed point in vertex-level truncations would each falsify the corresponding pillar. The negative results of Section 6 already falsified the easy versions of (ii); I regard that as progress.

## A Reproducibility

All computations use Python 3.12 (NumPy, SciPy, SymPy). The complete archive, including every script, figure, JSON result, the experiment ledger, and the controls that killed the false positives, is available at <https://github.com/infinition/it-from-fix>. Beta functions were transcribed from the LaTeX source of [5] and cross-checked against its analytic fixed-point formulas. Not every late-stage experiment retained its exact execution script; the repository documents precisely which results are byte-reproducible and which are reconstructions from logged rules and outputs.

## B Program lineage

This note is one branch of a research program running since 2024, and the version numbers of its branches are independent. For the record: LVS (Latent Vacuum Stationarity, 2024 to April 2026) formulated stationarity as a selection principle and tested it on Standard Model couplings, ending in the negative and inconclusive results summarized in Section 3.1. RAPC (April 2026) probed emergent locality from modular correlations through a sequence of falsification gates, whose negative reinforcement-flow results are cross-referenced in Section 6. A separate dossier titled It from Fix v0.7 (May 2026) develops a distinct effective hypothesis, a state-dependent bilocal correction to semiclassical gravity with an equivalence-principle test as its signature; despite sharing the It from Fix name, that line is not the subject of this note. The present note, the Kernel Principle branch, starts its own numbering at v0.1; this is v0.3. All branches, including retracted explorations and their controls, are preserved in the program archive.

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